MILITARY SPECIFICATION

BOILER FEEDWATER AND BOILER WATER EDTA HARDNESS TEST KIT AND COMPONENTS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a water hardness test kit for use in determining the amount of hardness in boiler feedwater and boiler water on Naval ships.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-300 - Chemicals, Liquid: Packaging and Packing of.
PPP-C-301 - Chemicals, Dry and Paste: Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION
Code of Federal Regulations, Title 49

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications and standards, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN CHEMICAL SOCIETY (ACS)
Specifications - Reagent Chemicals

(Application for copies should be addressed to Applied Publications, American Chemical Society, 1155 16th Street, NW, Washington, DC 20036.)

MANUFACTURING CHEMISTS ASSOCIATION, INC.
Manual L-1 - A Guide for the Preparation of Warning Labels for Hazardous Chemicals

(Application for copies should be addressed to the Manufacturing Chemists Association, Inc., 1825 Connecticut Avenue, NW, Washington, DC 20029.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Component parts. The water hardness test kit shall consist of apparatus and reagents as specified herein.

3.2.1 Apparatus. Water hardness test kits shall consist of automatic-zero burette titration assembly and two brass measuring dippers.

3.2.1.1 Automatic-zero burette titration assembly.

3.2.1.1.1 Automatic-zero burette. The automatic-zero burette shall have a capacity of 10 milliliters (mL). The burette shall be so constructed that solution in excess of that required to bring the level of liquid in the filled burette exactly to the zero mark will drain back into the titrant reservoir bottle. The burette shall be graduated in tenths of a mL, with the numbers placed to the side so that interference with taking readings by the inner feed-tube is avoided. The graduations shall be easily readable. The burette shall be made from a high grade of alkali-resistant glass tubing and shall have the following dimensions:

Feed-tube: 8 millimeters (mm) by 1 mm by 9 inches long
Inside siphone tube: 1 mm wall thickness by not more than 5 mm outside diameter
Discharge tube containing stopcock: 4 inches long
Vent hole: not more than 1 mm diameter
Length of 10 mL scale: 14 - 16 centimeters with 100 0.1 mL graduations
Stopcock: Teflon, with 1 mm opening in plug
Overall length of burette: 18 inches

The discharge tube shall be attached as near the bottom of the burette as possible to facilitate complete drainage and shall be directly in front of the scale graduations. The axis of the stopcock plug shall be set at an angle of approximately 15 degrees to the horizontal, with the handle up and to the right, and with a rubber washer on the lower end to prevent the plug from pulling out. The scale graduations shall end above the opening of the discharge tube such that there will be a clear length of approximately 1.5 inches between the bottom of the scale and the discharge tube. The burette shall deliver 10 $\pm$ 0.1 mL of solution. Subsequent to manufacture, the burette glass shall be annealed to remove all strains.

3.2.1.1.2 Titrant reservoir. The burette assembly shall be provided with a titrant reservoir in the form of a polyethylene squeeze bottle of approximately 500 mL capacity, a rubber insert plug and a polyethylene screw cap, both with openings to accommodate the burette feed-tube.

3.2.1.1.3 Titrant reservoir support stand. The burette assembly shall be provided with a titrant reservoir support stand of dense polyethylene, with an accommodation for the titrant reservoir bottle.

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3.2.1.2 Brass measuring dipper. The brass measuring dipper shall be approximately 4 inches in overall length, with a handle of approximately 3-3/4 inches in length. The dipper capacity shall be approximately 0.2 mL. The brass shall be of alkali-resistant composition.

3.2.2 Reagents. The hardness testing kit shall contain the following reagents:

Titrating solution
Buffer
Indicator
Dimethylglyoxime

Reagents shall be subjected to the test of 4.7.1.2 performed in accordance with 3.3.

3.2.2.1 Titrating solution. The hardness titrating solution shall consist of an aqueous solution of disodium ethylenediaminetetra acetate (Na_2EDTA) of concentration 1.00 mL = 1.00 milligrams (mg) $CaCO_3$ and of pH 10.5 $\pm$ 0.1.

3.2.2.2 Buffer. The hardness buffer shall be a dry powdered water soluble mixture. A specified amount of buffer when added to a specified volume of water sample shall maintain a pH of 10.5 + 0.1 in the sample during the entire course of the hardness test (see 4.7.1.2). The hardness buffer shall consist of a uniform mixture of:

	Percent
sodium sulfide	1 to 2
sodium hydroxide	1
potassium sodium tartrate	1
sodium tetraborate	4
magnesium disodium ethylenediaminetetra acetate	0.1
mannitol	93

All of the above reagents shall be in accordance with the American Chemical Society (ACS) grade.

3.2.2.3 Hardness indicator, ACS grade. The hardness indicator shall be a dry powdered water soluble mixture of Eriochrome Black T, color index 14645, dispersed in sugar, which at pH 10.5, shall impart a red color to a water sample containing hardness and a blue color when hardness is absent.

3.2.2.4 Dimethylglyoxime (DMG) powder, ACS grade. DMG, when added to a water sample containing sufficient dissolved nickel to interfere with the determination of an accurate titrimetric end point, shall eliminate the effect of the nickel on the end point.

3.3 Hardness test performance. When 100 mL of the standard 1.0 milliequivalents per liter (e/m) hardness solution is tested in accordance with the procedure of 4.7.1, it shall require not more than 5.1 mL nor less than 4.9 mL of the titrating solution.

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NOTE: The DMG is for use only in suppressing nickel interference in actual shipboard boiler feedwater samples. It shall not be used in performing the hardness test in this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article shall consist of the first complete hardness test kit produced under a contract (see 6.2.1). This test kit shall successfully pass the examination of 4.6 and the tests of 4.7, prior to beginning production. Any modification found necessary as a result of the inspection shall be incorporated into the remaining kits. The contractor shall prepare a first article test report in accordance with the data ordering document (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection (see 6.2.1) shall consist of the visual and dimensional examination of 4.6 and tests of 4.7 selected in accordance with table I from each inspection lot (see 4.5.1).

TABLE I. Sampling for inspection.

Number of kits in inspection lot	Number of kits in sample	Acceptance number (defectives)	Rejection number (defectives)
25 or under	5	0	1
26 - 50	8	0	1
51 - 90	13	1	2
91 - 150	20	1	2
151 - 280	32	2	3
281 - 500	50	3	4
501 - 1200	80	5	6

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4.5 Sampling.

4.5.1 Inspection lot. For purposes of sampling, a lot shall consist of all complete test kits presented at one time.

4.5.2 Sampling for inspection. A random sample of complete test kits shall be selected in accordance with table I and special inspection level S-1 of MIL-STD-105 from each lot presented for inspection and tests. AQL shall be 2.5 percent defective.

4.6 Examination.

4.6.1 Visual and dimensional examination. Each of the complete hardness test kits selected in accordance with table I shall be visually examined to verify compliance with this specification. If any complete test kit in the sample contains one or more defects, this shall be cause for rejection of the kit, and if the number of defective test kits in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.6.1.1 Apparatus. Items of apparatus shall be visually and dimensionally examined.

4.7 Tests.

4.7.1 Hardness test performance.

4.7.1.1 Standard hardness solution 1.0 e/m hardness. A standard hardness solution shall be prepared, containing 0.0555 $\pm$ .005 grams of dissolved calcium chloride per liter (L) of demineralized water. ACS reagent grade calcium chloride shall be used. An analysis of the hardness standard shall be performed using an accepted chemical method other than the hardness method herein specified. Identification of the method used and a certificate of analysis shall be made available to NAVSEA on request, and shall be included in the first article test report (see 6.2.2).

4.7.1.2 Hardness test. The hardness test shall consist of the following steps:

(a) Rinse the titrant reservoir bottle with two small portions of hardness titrating solution.
(b) Fill the titrant reservoir bottle with titrating solution as supplied. No dilution is necessary.
(c) Slide the rubber insert over the barrel of the burette.
(d) Screw the burette assembly firmly onto the titrant reservoir bottle.
(e) Fill the burette by squeezing the bottle. Allow to drain for flushing.
(f) Refill the burette and ensure that air bubbles are absent in the burette tip.

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NOTE: Keep the dippers for buffer and indicator dry when not in use. Do not interchange the dippers when running a series of hardness tests, because this will result in contaminating the buffer with the indicator and vice versa. Dippers should be rinsed with distilled water and shaken dry when testing is completed.

(g) Rinse a porcelain casserole and stirring rod with distilled water.
(h) Rinse a 100 mL graduated cylinder with distilled water and then with a portion of the 1.0 e/m standard solution. Measure 100 mL of 1.0 e/m standard solution into the graduated cylinder and pour into the casserole.
(i) Add 2 level dippers of hardness buffer and stir until dissolved.
(j) Add 2 level dippers of hardness indicator and stir until dissolved.
(k) The 1.0 e/m standard solution will be red. Slowly add titrating solution dropwise, with stirring, to a blue end point.
(l) Read the burette to the nearest 0.1 mL, record the burette reading, multiply the reading by 0.2 and record the result in e/m hardness.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition.)

5.1 Domestic shipment and early equipment use.

5.1.1 Packaging.

5.1.1.1 Levels A and C.

5.1.1.1.1 Titrating solution. The titrating solution shall be packaged in 1 gallon polyethylene bottles, conforming to type I, class 1 of PPP-C-300 for bottles and closures. Closures shall be dense polyethylene, continuous-thread screw-type caps making an air-tight seal with the bottle.

5.1.1.1.2 Hardness indicator. The hardness indicator powder shall be packaged in 100 gram quantities in wide-mouth, low actinic glass bottles, conforming to type I, class 1 of PPP-C-301 for bottles and closures. Closures shall be dense polyethylene, continuous-thread screw-type caps making an air-tight seal with the bottle.

5.1.1.1.3 Hardness buffer. The hardness buffer powder shall be packaged in 100 gram quantities in wide-mouth, low actinic glass bottles, conforming to type I, class 1 of PPP-C-301 for bottles and closures. Closures shall be dense polyethylene, continuous-thread screw-type caps making an air-tight seal with the bottle.

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5.1.1.1.4 Dimethylglyoxime. The DMG powder shall be packaged in 1 ounce quantities in wide-mouth, low actinic glass bottles, conforming to type I, class 1 of PPP-C-301 for bottles and closures. Closures shall be dense polyethylene, continuous-thread screw-type caps making an air-tight seal with the bottle.

5.1.1.1.5 Compatibility of container. The container, closure, lining, or space fillers shall not interact physically or chemically with the contents of the container so as to be altered or to alter the strength, quality or purity of the container contents for at least 24 months from the date of shipment.

5.1.1.2 Commercial.

5.1.1.2.1 Automatic-zero burette titration assembly and brass dipper. The packaging of the automatic-zero burette titration assembly and brass dipper (2 per kit), which may be the contractor's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use.

5.1.2 Packing. Hardness testing reagents, packaged as specified in 5.1.1, shall be packed level A, B, or C, as specified (see 6.2.1) in accordance with PPP-C-300, or PPP-C-301, as applicable. Items of apparatus (burette assembly, dippers) shall be packed in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early use. The shipping containers or method of packaging shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with MIL-STD-129. The information shall include nomenclature, National stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.3.1 Special marking.

5.1.3.1.1 Bottles.

5.1.3.1.1.1 Titrating solution. Each bottle shall be labeled as follows: "Hardness Titrating Solution 1.00 mL = 1.00 mg $CaCO_3$".

5.1.3.1.1.2 Hardness indicator. Each bottle shall be labeled as follows: "Hardness Indicator".

5.1.3.1.1.3 Hardness buffer. Each bottle shall be labeled as follows: "Hardness Buffer".

5.1.3.1.1.4 Dimethylglyoxime. Each bottle shall be labeled as follows: "Dimethylglyoxime Powder Meets ACS Specifications".

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5.1.3.1.2 Packages and shipping containers.

5.1.3.1.2.1 Automatic-zero burette titration assembly. Each package and shipping container shall be marked as follows: "Automatic-Zero Burette Assembly".

5.1.3.1.2.2 Brass dipper. Each package and shipping container shall be marked as follows: "Brass dipper, 2".

5.1.3.1.2.3 Titrating solution, hardness buffer and hardness indicator. Each package and shipping container shall be marked in accordance with 5.1.3 and shall include the following:

(a) Quantity
(b) Date of manufacture, month and year
(c) Expiration date of 24 months from date of manufacture (For example, manufactured 1/81, expires 1/83)

NOTE: "Mandatory shelf-life material. Discard material after expiration date".

(There is no expiration date for DMG.)

5.1.3.2 Hazardous chemicals. Package units of hazardous chemicals shall have warning labels as may be required by Code of Federal Regulations, Title 49, Parts 100-199, the Manufacturing Chemists Association Manual L-1, or Department of Defense instructions which shall take precedence.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of packaging, packing and marking for shipment shall be specified by the contracting activity (see 6.2.1).

6. NOTES

6.1 Intended use. This test kit is to be used for the determination of hardness in boiler feedwater and boiler water.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article tests are required for inspection and approval (see 4.3).
(c) Whether quality conformance tests are required (see 4.4).
(d) Preservation, packaging, packing and marking requirements other than those required (see 5.1 and 5.2).
(e) Whether the complete test kit or component part is required. Component parts are listed in section 3 of this specification.

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3 and 4.7.1.1	Reports, first article test	UDI-T-23790	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 Bid data. Bidders offering individual components rather than complete kits will be subject to the following special requirements:

(a) Bidders offering items of apparatus only will not be required to perform the tests of 4.7.1 and 4.7.1.2, but will be required to fulfill all other requirements of this specification. These bidders may request that the Government waive the requirements for first article samples, provided the bidder is offering a product which has previously been acquired or tested by the Government, and that bidders offering such products, who wish to rely on such productions or tests must furnish evidence with the bid that prior Government approval is presently appropriate for the pending acquisition.

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(b) Bidders offering reagent items only (titrating solution, buffer, indicator) will be required to fulfill all the requirements and tests of this specification. These bidders may request that the Government waive the requirements for first article samples, provided the bidder is offering a product which has previously been acquired or tested by the Government, and that bidders offering such products, who wish to rely on such productions or tests must furnish evidence with the bid that prior Government approval is presently appropriate for the pending acquisition.

Preparing activity:
Navy - SH
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